

Building a Unified Science of Natural Behavior with Densely Overlapping Measurement Environments [DOMES]

FreeMoCap Foundation, Inc.

Written Proposal to the NSF X-Labs Initiative

1. MISSION

Develop and disseminate a new class of scientific instrument called a **Densely Overlapping Measurement Environment (DOME)** that records the complete perception–action loop of a behaving agent as a single calibrated, uncertainty-tagged measurement, and build the autonomous organization dedicated to the measurement itself to align perceptual and motor neuroscience, musculoskeletal biomechanics, mobile robotics, and embodied AI into a single convergent science of natural behavior.

A New Observable — The Sensorimotor Loop. Every agent, living or engineered, must solve the same problem - sense a thin slice of available energy in the environment through a limited set of imperfect transducers and on its basis press against the world to propel itself in the direction of its goals. Information flows in, forces flow out; the brain exists to yank the bones around.

We can measure individual components of this loop with extraordinary precision progressing at the rate of our technology. Camera and IMU based motion capture can track the kinematics of the body, force plates measure the kinetic forces between the body and world. Outward facing cameras can measure the world, and inward facing cameras can track the eyes. For humans, EEG nets and surface EMG record coarse records of patterns of neural activity in the cortex and musculoskeletal system. In animal models, dense electrode arrays resolve the same signals to the precision of individual spikes of neurons and motor units.

And yet, despite the extraordinary precision and progress along these individual threads of measurement, access to the coherent whole representing the shared observational context that each of these tools illuminates remains elusive. Occasionally, a research team will reach across disciplinary divides and stitch several threads together for heroic integrative studies, but the methods rarely escape the Methods sections of a small cluster of publications spanning roughly the timescale of a single PhD.

We need a new organization that exists outside of the siloed domains of the Academia that dedicates itself to the MEASUREMENT, the development and dissemination of world-class, convivial tools and novel skillsets to explore the landscape the capacity unlocks. That organization is the **FreeMoCap X-Lab [FMC-X]**. We have already changed the landscape human-focused research by bringing high-quality motion capture to a global community. With the support of the NSF X-Labs program, we will extend our mission and change the face of science.

A New Instrument — Building the DOME. A DOME is a densely instrumented region of real space engineered to (at some aspirational extreme) record every measurable channel of the agent–environment interaction. In practice, a given DOME carries only a subset of the available instrumentation, yet all DOMES produce the same output. A DOME is defined by the ontology of its domain rather than its current instruments. A metrologically-grounded record of canonical models defined by their degrees of freedom, which maybe unspiced or hydrated by measurements from whatever instruments are present. A reconstructed Skull is assumed to carry Eyes whether or not an eye tracker measured them, and a RetinalInput or RigidBody

denotes the same quantity whether it came from a human, a ferret, or a marmoset (See § 3 and Table 2).

The New Capability. The same **Sensor-Grounded Ontology** that lets heterogeneous DOMES express their measurements in one shared schema makes the instrument legible to a human — explicit entities, formalized relationships, and explicit chains of reasoning from sensors to canonical objects — also makes natively legible to a machine: typed, grounded data over which a model can build reliable reasoning chains rather than guessing at unlabeled streams. By treating complexity management as a first-order concern; a DOME gains a clarity of expression that lets it route its live streams and logs through a local model to assist novices, experts, and developers alike, and that lets published results tag back into one shared entity-relationship structure connecting them to the wider body of science. In its own.

Unmet Needs. Building a scientific instrument is a distinct craft, not a byproduct of domain expertise. Integrated measurement is repeatedly scoped as a student side-project and abandoned when it proves to require a career of engineering rather than a semester of it, and even its successes rarely outlive the trainee who built them. By Conway's Law, an enterprise of domain-named departments, journals, and grants tends to produce only domain-scoped instruments — never the sustained metrology that turns a proof of concept into shared infrastructure. A tool defined by a measurement rather than a domain unites everyone who needs it and becomes a commons precisely because it imposes none of their goals. The measurement therefore needs its own organization, scoped to the instrument and staffed to maintain it for years — what the X-Labs model exists to fund and a university appointment cannot sustain.

The Vision. The vision of this proposal represents a grand acceleration of movement the FreeMoCap Project has fostered since 2021, and there is no upper limit to our ambition. The global spread of the FreeMoCap software shows that the world has a hunger for this kind of tool, and we intend to feed it. We will build the instrument that will unlock that ability to ask an unknowably vast array of questions, and those questions will carry our tools to every corner of human curiosity. There will be a DOME in every research lab, every classroom, every Physical Therapy clinic, animation studio, and athletic facility. Field-deployable wearable DOME-MOBILE systems that provide lab-quality data of humans and animals in the remote wilderness. They will be built into the SCUBA systems of deep sea welders and into the EVA suits and HAB volumes of the ISS. Every country hospital and retirement home will have an AI-enabled, auto-calibrating, self-analyzing Gait and Posture facility that will have plain language conversations with patients about their own kinematic data. There will be DOMES bolted into the sides of mountains to measure the aerodynamics of hunting hawks, and onto the hulls of shipping boats to understand the mechanics of the dolphins that play in their wake. Every exotic animal enclosure in every zoo could become an automated animal behavior lab, managed through citizen science and then accelerated through auto-training backprojection pipelines.

The Internet did not reshape the world by making computer run faster, it did so by giving them the ability to communicate. In doing, it turned a disparate array of powerful machines, into an interoperate network that changed what it means to be a Human.

We will build the DOME - The opposite of a space telescope. An empty volume of lenses focused inward on where the body meets the world. We will build a new observable and share it broadly to anyone that wants it. We will change our understanding of ourselves, and the ways we interact with the world.

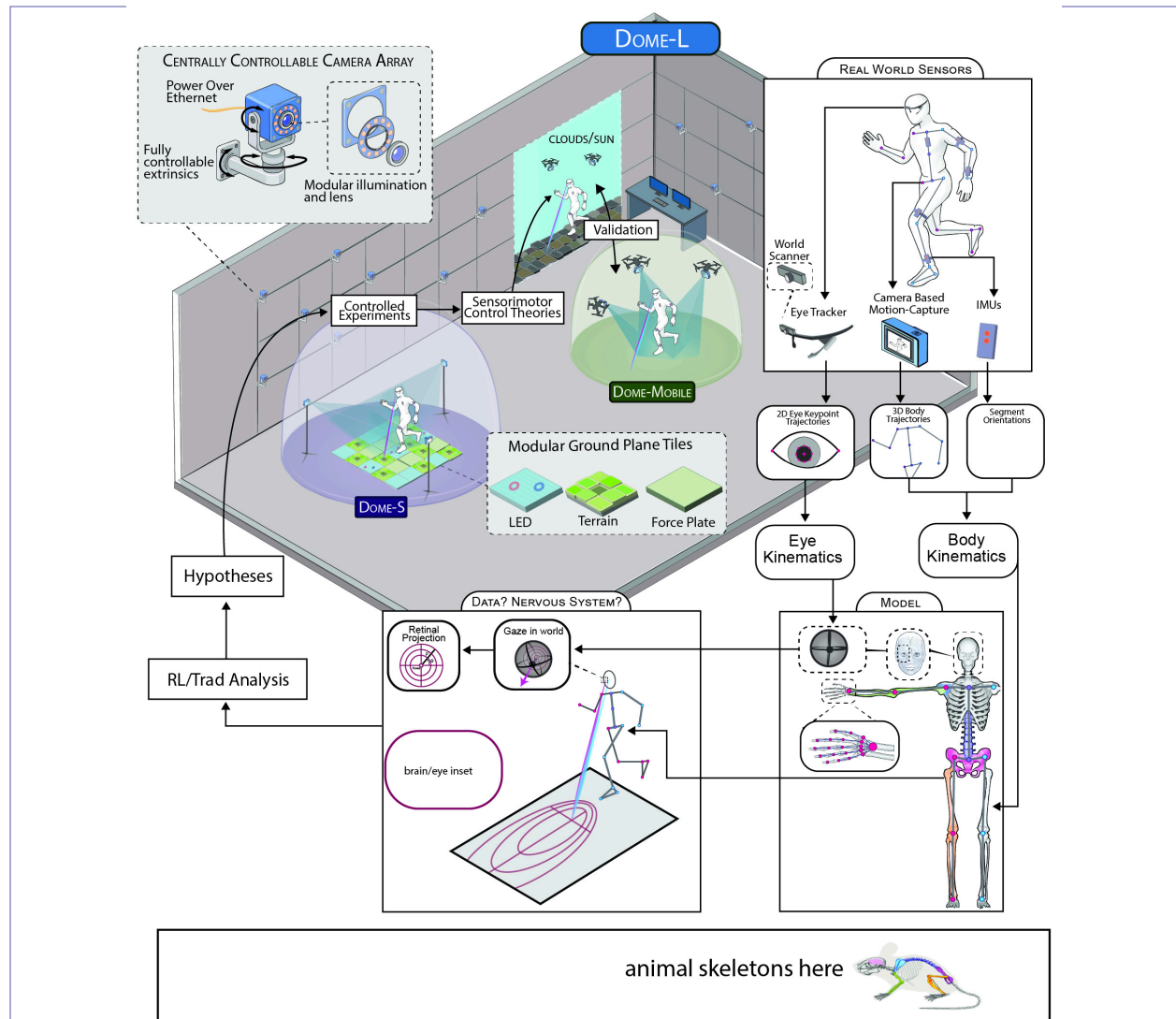


Fig.1: A Densely Overlapping Measurement Environment (DOME). The flagship DOME-L physically contains the disseminated DOME-S and wearable DOME-MOBILE forms, calibrating every variant against one reference. A centrally controllable camera array and modular force/LED/terrain floor tiles feed calibrated eye, body, world, and force streams into a single model of the sensorimotor loop – segment kinematics, gaze-in-world, and reconstructed retinal input – closing an analysis-to-hypothesis-to-experiment loop inside the same instrument that measures it.

2. TECHNOLOGY LANDSCAPE

The DOME is organized around a single, demanding measurement target: reconstructing the visual input to a behaving agent's retina precisely enough to predict the neural activity it drives. Concretely, that means estimating gaze-in-world to better than 1° – roughly the diameter of the human fovea – and resolving eye kinematics finely enough to recover retinal optic flow and its derivatives, divergence and curl ($^\circ$; Fig. 2). Reaching that target decomposes into three measurement threads, each of which must hit its own precision and then compose into one calibrated record: the **body** that carries the eye, the **eye** itself, and the **world** it looks at. The state of the art, and the gap the DOME closes, is clearest thread by thread.

The body. Body kinematics are captured today by markerless pose estimation (e.g. FreeMoCap, riding a decade of computer-vision advances – OpenPose, DeepLabCut, MediaPipe), marker-based systems (Vicon, Qualisys, OptiTrack), and inertial (IMU) suits (Xsens), with the forces behind that motion measured by force plates (AMTI, Bertec),

instrumented treadmills, and pressure insoles. Markerless capture already reaches 2–6° joint-angle accuracy against research-grade references, but in practice most users achieve far less accuracy due to errors in camera set up and calibration.

We target that gap with a **self-calibrating, actuated camera array** capable of tuning its own extrinsics and intrinsics to optimally cover a selected capture volume, allowing for fluid centralized control and automated AI-self-run experiments on best practices for camera arrangement during motion capture. We use this capacity to validate smaller camera-based capture volumes and create sensor-fused estimates that combine triangulated keypoint trajectories with IMU-suit based segment orientation estimates to generate **reproducibly** high quality kinematic estimations from any measurement system — both within the lab and out in the world.

The eye. Sub-degree gaze-in-world may be reachable with commercial instruments plus FreeMoCap: the best mobile trackers reach 0.6–1.8° in the head frame (Tobii Pro Glasses 3, Pupil Labs Neon, AdHawk MindLink), and the only published end-to-end eye + IMU + world reconstruction — this team’s own prior work — composes them to $\pm 1^\circ$ calibrated and $\pm 2\text{--}3^\circ$ in natural walking (^{2,3}). That much is careful extension of prior art. Retinal **curl**, though, is beyond every available tool, because no eye tracker measures ocular torsion. Vision science has long set torsion aside — defensibly, since it could not be measured and since Listing’s Law specifies eye orientation from just two of its three degrees of freedom (elevation and abduction, neglecting torsion). But Listing’s Law fails whenever the vestibulo-ocular reflex is active — that is, during essentially all natural behavior — and¹ implicates retinal curl as a likely control signal for real-world locomotion, a signal the torsional axis directly shapes. Measuring that missing degree of freedom is the design target of our next-generation, dense-sensor-array eye tracker.

The world. Reconstructing what lands on the retina requires reconstructing the scene the eye looks at. Scene geometry is captured today by photogrammetry, Gaussian splatting, RGB-D, and LiDAR. The team’s own photogrammetry recovers natural terrain to a repeatability near 3 cm — about 0.5° of visual angle. But these scanners are passive and stand outside the agent: none is fused into the moving observer’s coordinate frame with propagated uncertainty. We mount the scanner on the agent — a dense-array world-scanner carried on the head — and extend it outdoors on the same drone-swarm nodes that carry the body cameras, so the scene is measured in the same calibrated frame as the body and eye. Inside the flagship, an actuated floor of force, LED, and terrain tiles turns the ground itself into a controllable experimental stimulus (Fig. 1).

The gap is integration, not any single sensor. Each thread is excellent and improving on its own, but without an integration they are evolving away from each other. Two adjacent fields are stalled for want of exactly this record. Computer-vision pose estimation has drifted from the pixels toward ungroundable targets — 3D-from-2D, mesh fitting, semantic segmentation — while still benchmarking against COCO imagery scraped from the early 2000s, and it fails on the out-of-distribution movement that matters most: clinical gait, extreme athletics, animal locomotion. It needs physical ground truth at scale, which a calibrated multi-camera DOME produces by construction. Legged-robot reinforcement learning is limited not by simulation — MuJoCo, Isaac Lab, and Genesis are excellent — but by the sim-to-real gap, and the resource that would close it is uncertainty-tagged, multimodal recordings of real agents in real environments; the standard corpora (Human3.6M, AMASS, GRAB) are single-modality, lab-constrained, or carry no uncertainty at all. The same DOME record feeds both.

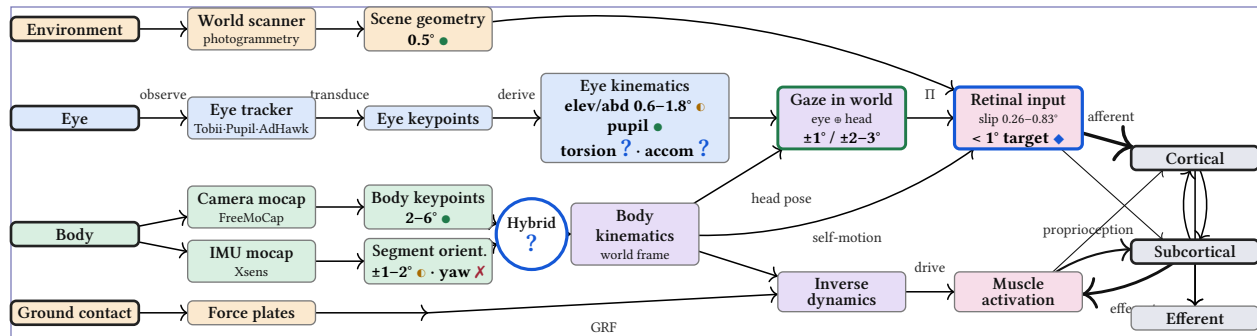


Fig.2: Quantifying the technology landscape. Reported error sits on the **quantity measured**, not the instrument, tagged by provenance: ● open and traceable, ○ closed vendor model (uncertainty unknown), ✗ closed and **not truth-preserving** (Xsens fills an out-of-bounds magnetometer yaw with a model fit), ◆ DOME target, and a blue ? for what no instrument measures yet. Mobile eye trackers resolve head-frame gaze to 0.6° (Tobii Pro Glasses 3), ≈1° (AdHawk MindLink), and 1.3–1.8° (Pupil Labs Neon), but none measure ocular torsion or accommodation, and the camera–IMU fusion is itself unbuilt. Composed into gaze **in the world** through IMU segment orientation (±1–2°), markerless keypoints (2–6°⁴), and photogrammetric scene geometry (≈3 cm ≈ 0.5°), the only published end-to-end eye + IMU + world method — this team’s own prior work — reaches ±1° calibrated and ±2–3° in natural walking (^{2,3}), against the sub-1° needed to predict the neural activity it drives. No existing toolchain delivers sub-1° gaze-in-world with a propagated uncertainty budget — the DOME goal.

3. OUTCOMES

The instrument. Our five-to-seven-year outcome is a single instrument — the DOME — realized in three human-scale forms and an animal-scale form that records the complete perception–action loop as one calibrated, uncertainty-tagged measurement and yields the same canonical record regardless of which sensors are present. DOME-L is the warehouse-scale flagship in Greater Boston; it drives each sub-instrument to its limit and is large enough to physically contain the smaller forms, which makes it the metrological reference for the entire network. DOME-S extends FreeMoCap’s webcam capture into the disseminated lab- and classroom-scale form, and DOME-Mobile is the wearable form that carries the measurement onto natural terrain outside the lab. Because the instrument is defined by its measurement ontology rather than by any one sensor, a markerless-camera estimate and an inertial estimate describe the same kinematics and fuse into a single estimate better than either alone.

The organizing benchmark - the retinal projection. Every outcome below is a term in one uncertainty budget: reconstructing veridical retinal input — the world projected onto a moving agent’s retina — to within 1° at the fovea while the agent fixates a point on the ground ahead. That target, and the current-art error sources it composes, are established in the Technology Landscape (§ 2; Fig. 2); here each subsystem carries its own quantitative sub-target beneath it.

Body — A self-calibrating, semi-autonomous camera array. On the motor half of the loop we deliver kinematics clean enough for inverse dynamics from markerless capture. The actuated, self-calibrating camera array places daisy-chained Power-over-Ethernet cameras on programmable mounts that stay metrically calibrated as one instrument while being re-aimed and re-focused from a single console, dissolving the practical bottleneck that tends to leave large motion capture rigs frozen in place due to the high cost of manual re-calibration. This is adaptive AI-based sensing in hardware: the algorithm reconfigures the physical optics to maximize observability, running autonomous experiments over camera geometry rather than settling for a hand-placed rig. Success is measured as post-reconfiguration calibration quality and the time from sub-volume selection to calibrated capture, against a fixed-calibration reprojection baseline. Camera-derived keypoints and IMU-derived segment orientation estimates are fused then combines noisy-but-accurate outside-in estimates with precise-but-drifting inside-out estimates under explicit per-joint uncertainty, validated as joint-torque

uncertainty against force-plate ground truth and muscle-force uncertainty against EMG-informed estimates.

The starting point is already quantified: FreeMoCap recovers joint centers to under 30 mm and sagittal joint angles to 2–4° RMSE against research-grade optical motion capture during treadmill gait, from a roughly \$120 six-webcam rig⁴. The actuated array and camera-IMU fusion target the residual gap to inverse-dynamics-grade kinematics — per-joint uncertainty tight enough that derived joint torques and muscle forces carry meaningful error bars.

Eye — the next generation of eye trackers. Mobile eye tracking has been stalled for at least a decade. There are more powerful arrays of multispectral optical sensors on the back of a cell phone than there are in a modern eye tracker. We will build a new kind of eye tracker that measures degrees of freedom commercial trackers omit, with spatial and temporal accuracy beyond current expectation. We will design a dense array of multi-spectral sensors (including high-resolution **stereo RGB-IR sensors, ultra-high speed event cameras, 3d scanning LiDAR, and single-pixel MEMS imagers**) to reconstruct the eye to a level of spatiotemporal precision beyond the capacity of any existing system. We will keep our unit costs low by taking advantage of the economics of the commodity-imager curve, with the goal of creating a cost-collapse for next-gen eye tracking the way FreeMoCap did for motion capture.

The new eye tracker will resolve ocular torsion from iris texture — the degree of freedom that corrupts the curl of retinal optic flow and a standing research target¹. Realtime and post-hoc pipelines will produce microsaccade-scale resolution, and possibly unlock estimates of lens accommodation by tracking the dual-Purkinje image. Its benchmarks are sub-15-arcminute eye-in-head accuracy, 1° torsion accuracy, and the composed 1° gaze-in-world target above.

For context, the best shipping mobile trackers report 0.6° (Tobii Pro Glasses 3), 1° (AdHawk MindLink), and 1.3–1.8° (Pupil Labs Neon) gaze accuracy in the head frame — and none of them measure ocular torsion or lens accommodation at all (Fig. 2). The sub-15-arcminute (0.25°) eye-in-head target is finer than any of these, and the torsion and accommodation channels are new measurements rather than incremental ones.

World — the scene the agent acts in. Reconstructing what the eye sees requires reconstructing the world it acts in, so a head-mounted world-scanner of stereo-RGB, structured-IR, and LiDAR produces per-frame RGB-D and, over longer horizons, a terrain mesh and a SLAM-corrected head trajectory that fixes inertial drift³. The same scanner package mounts on drone-swarm nodes that act as a mobile extension of the camera array, grounding DOME-Mobile’s inside-out estimates outdoors where a fixed rig cannot reach, benchmarked as gaze and kinematic uncertainty against the DOME-L reference during co-recorded trials. Modular floor panels — each a force plate, terrain tile, or LED surface — and the ARGP perturbation platform close the world half of the loop by making the ground itself both sensor and controllable stimulus.

One record across species. Functionally equivalent instruments at collaborator labs yield the same canonical records as the human instrument. The animal-scale instrument already exists — the team’s ferret and mouse builds integrate three-camera gaze, full-body kinematics, and world cameras in one calibrated system, with neural recordings being added now — and the human, mouse, and ferret builds are being brought to functional equivalence, extending to guinea fowl for musculoskeletal biomechanics with EMG MD and marmoset for primate electrophysiology AH/JY. Success is cross-species measurement comparability under shared calibration and data-exchange protocols, so that a reconstructed RetinalInput or RigidBody denotes the same thing whether it came from a human, a ferret, or a marmoset.

What the record unlocks. Because every channel is calibrated, synchronized, and uncertainty-traced within one shared sensor-grounded ontology, a DOME instrument generates its own training signal by back-projecting calibrated 3D reconstructions onto each camera's 2D view, and the resulting reprojection error fine-tunes pose estimators, measured as improvement on out-of-distribution movement — clinical gait, acrobatics, animal locomotion — rather than on a saturated benchmark: an instrument engineered for next-generation AI training pipelines. The record is natively reinforcement-learning-ready because a DOME export maps directly onto MuJoCo and Isaac Lab rigid-body models with uncertainty intact, so DOME-recorded corpora drive sim-to-real policy transfer and inverse-RL recovery of the control policies underlying observed behavior, testable back in DOME-L via ARGP perturbation.

This self-improvement loop is already demonstrated at small scale: a generic pose estimator mistracks a transfemoral prosthesis by up to 152 mm at the ankle, but training a task-specific keypoint model on the instrument's own calibrated data drops that error to 8–14 mm⁴. What was a one-off dissertation result becomes, in the DOME, a repeatable methodology capable of driving measurement beyond the capacities of the present landscape.

Dissemination and validation. The outcome is a countable, validated footprint rather than a released codebase: [N] validated DOME-S instances spanning human biomechanics, robotics, visual neuroscience, and clinical sites, each cross-validated against the DOME-L reference and shipped with a build guide, parts list, and calibration protocol any lab can follow. We anchor clinical rigor to FDA 510(k) clearance of the FreeMoCap/DOME instrument for rehabilitation and assessment, letting a regulatory bar drive protocol quality. Phase activities and maturity targets for every line above appear in the Milestone Matrix (Table 1), whose quantitative exit criteria are the benchmarks named here. [TODO - Are they? Make double sure about that...]

Item	In place	Phase 0 · 9–12 mo	Phase 1 · 24–36 mo	Phase 2 · variable
ORGANIZATION — OPERATIONAL DELIVERABLES				
Legal & governance	✓ 501(c)(3)	Governance + Autonomy Plan	board seated	
Facility (lease)		lease + move-in	office fit-out	
Full-time team	✓ core FT	hire · onboard	full staff	
Required NSF plans		Data · IP · Security · Gov; Ph1 plan + budget		
IP management	✓ prior IP	IP Mgmt Plan	license · manage	
DOME VARIANTS				
DOME-L (Automated)		PoC / testbed	Prototype - Validated	Operational
DOME-S (Static)	✓ FMC webcam	Validated	Operational (510(k) cert)	Operational
DOME-MOBILE		PoC-Breadboard	Prototype-Validated	Operational
BODY — TECHNICAL DEVELOPMENT				
Camera array		PoC-Breadboard	Prototype -> Validated	Operational
IMU↔camera fusion	✓ FMC mocap	PoC	Prototype	Validated
EYE — TECHNICAL MATURITY				
Dense Array Eye tracker		PoC - Breadboard	Prototype-Validated	Operational
WORLD — TECHNICAL MATURITY				
Dense Array World-scanner		PoC - Breadboard	Prototype-Validated	
Actuated floor (force-LED-terrain)	✓ v1	PoC	Prototype	Validated
Drone Swarm Mocap			Concept-PoC	PoC-Validated

Table1: Maturity ladder (TRL): Concept (2) · PoC (3) · Breadboard (4) · Prototype (5–6) · Validated (7) · Operational (8–9); organization rows show operational deliverables. ✓ = in place pre-award. Each Phase 1 Progress Milestone carries a quantitative exit criterion (FOA §9.2).

4. SENIOR/KEY PERSONNEL

Under the NSF X-Labs model, Phase 0 is the runway for members to transition to full-time before Phase 1. The PI and CSO are already full-time on FreeMoCap; the others commit to full-time by Phase 1.

Jonathan Matthis, PhD — President / PI. Founded and maintains FreeMoCap; two decades measuring the full perception–action loop. Left a tenure-track professorship (Human Movement Neuroscience, Northeastern) because it could not support the integrated tool-

building this work requires. NEI NIH K99/R00 recipient (EY028229), with a publication record spanning the full loop, from outdoor gaze-and-gait to real-world retinal-input reconstruction.

NR — Chief Executive Officer. Technology-sector engineering leader and startup co-founder, recruited to run the organization so the PI can focus on the technical Mission. Staff Platform Software Engineer at Rapid7; co-founded and scaled MailLift, a venture-backed API startup. Brings the leadership and operations capacity to build FreeMoCap into an autonomous, independent X-Lab.

EI — Chief Technology Officer. Enterprise software architect (20 years experience in Tech industry, engineer to Principal Architect / CTO) and FreeMoCap's CTO since 2021, essentially since founding. Concurrent CTO at Chorus Innovations; former Principal Architect at Unqork; distributed-systems and data-engineering depth (Python, Node.js, PostgreSQL, Spark; RDF/ SPARQL). Ensures enterprise-grade architecture and the calibrated, semantically-unified data backbone the instrument requires. Represents a skillset not present in academic science.

AC, PhD — Chief Scientific Officer. Dissertation validated FreeMoCap against research-grade optical motion capture (Qualisys), demonstrating clinically valid kinematics from commodity hardware. Expertise in clinical tool validation, bench-to-bedside development, neuroprosthetics, and brain imaging; leads DOME-S development and clinical dissemination.

JKL, PhD — Chief AI Officer. Cognitive scientist and applied-AI architect (Applied AI Architect, SOLID Inc.); FreeMoCap contributor since 2024 and co-developer of SkellyBot, an AI-based teaching assistant that he and the PI build and deployed into 10 classes across 3 universities between Spring 2023 and Fall 2025. 15+ years handling sensitive human-subjects and health data under FERPA and IRB — anchors data governance and research security, and builds internal AI for development, support, and education.

RR — Chief Financial Officer. Cooperative- and nonprofit-finance specialist and FreeMoCap's Treasurer / volunteer CFO since 2021, its longest-serving financial officer. Founder of Capital Bookkeeping Cooperative; Controller at Honest Weight Food Co-op; Adjunct Professor of Accounting, University at Albany (MS, Accounting). Runs financial operations and grant management; cooperative-governance experience fits the Foundation's open structure and the X-Labs autonomy requirements.

KM, PhD — Project Manager, DOME-Mobile. Computer-vision engineer, data scientist, and longtime PI co-author whose work maps almost one-to-one onto DOME-Mobile. Built the outdoor eye-tracking + photogrammetry + IMU pipeline the DOME generalizes^{3,5}; currently Senior Data Scientist at SynMax (multi-sensor fusion, cloud MLOps). Holds essentially the complete skillset for DOME-Mobile's core instrumentation.

MN — Project Manager, DOME-L. Clinical biomechanics, medical-systems design, mechatronics, and multi-modal mocap-lab management; holds the skillset for large-scale DOME-L facility design and operation.

5. TEAM CAPABILITIES STATEMENT

Building the DOME demands a combination of expertise that no single institution's faculty could assemble, because the work is neither pure science nor pure engineering but the sustained craft of turning one into the other. The FreeMoCap Foundation was built from the ground up to hold that combination under one roof. The leadership team pairs scientific domain expertise in perception-action neuroscience and clinical biomechanics with enterprise software architecture, applied-AI systems, and hardware and mechatronics engineering at both wearable and facility scale, all run by dedicated operations and finance leadership recruited so the technical staff can stay on the Mission. Each member is detailed individually in § 4; what

matters here is that they already function as a single team, having built and maintained a shipping instrument together — most of them for years, part-time or volunteer alongside senior industry and academic roles, before any funding existed.

Governance and autonomy. The FreeMoCap X-Lab (FMC-X) will operate as an independent project within the FreeMoCap Foundation, a 501(c)(3) nonprofit whose leadership is nearly identical to FMC-X's and whose existing mission almost fully overlaps it. Final decision authority rests with the President/PI at the top of the org chart, so research, partnership, and hiring decisions are made in days rather than weeks, with no higher management to consult. The Foundation is already autonomous as of this submission: it holds full internal control of funding allocation, research direction, partnership agreements, intellectual property, and hiring, requires no parent-institution approval for any operational decision, and satisfies every condition of the NSF X-Labs Autonomy Factor Test (§6.1.1) at the time of submission. Phase 0 is the runway over which the remaining leaders transition to full-time ahead of Phase 1; the PI and CSO already are.

Network, partnerships, and community. A standing, active network of research groups already spans every domain the DOME serves — human perception-and-action researchers, robotics and prosthetics groups, visual neuroscientists working in ferret and mouse, musculoskeletal biomechanists working in guinea fowl, and primate electrophysiologists (Table 2).

Initial	Institution	Model
BS	CU Anschutz	Ferret
MD	UC Irvine	Guinea fowl
DF	CCMH	Mouse
AH	UCLA	Marmoset, NHP
MH	UT Austin	Human
KB	Indiana Univ.	Human, NHP
JY	UC Berkeley	Human, NHP
GN	RAI Institute	Robotics
CH	FAMU-FSU	Robotics
JH	Agility Robotics	Robotics
SH	Cornell	Robotics

Table 2: Collaborator network.

Under X-Lab funding, we will merge this collaborator network and the globally distributed community FreeMoCap users to form a coherent culture organized around a shared set of tools, skills, and measurements. We will communicate design plans and roadmap priorities through structured processes that gather broad input without stalling on consensus — a Request-for-Comments process modeled on Python's PEP system.

Once a year, we will host an overlapping set of in-person events including workshops that train students and out-of-domain experts, hackathons that onboard developers, conferences that share DOME-based research, and a congress that sets organizational direction. A Community Grants Program and Developer Fund sustain the open-source network and double as a testbed for community engagement and consensus-seeking mechanisms.

A proven model. FreeMoCap has already proven this operating model. It is an open-source markerless motion-capture instrument, built and maintained outside academia, scoped to the measurement itself with an obsessive focus on usability and refusing to claim any research domain — and that refusal is precisely what let it become a commons rather than one lab's tool. The receipts are concrete: over 15,000 users across 152 countries, more than 10,000 GitHub stars, and over 3,500 Discord members. The DOME applies the same measurement-scoped, master-built, open, full-time-maintained model to the far larger target of the complete interaction loop — a scale reachable only with the dedicated, autonomous, milestone-based funding the X-Labs program exists to provide.

Intellectual property and dissemination. The IP strategy keeps the core instrumentation open while actively pursuing adoption and commercialization: an open-source core under permissive licensing for downstream use, structured so the platform technologies stay widely accessible as the team pursues adoption, commercialization, and ecosystem growth. A full IP

Management Plan is due by the end of Phase 0 under the OT agreement and will be developed against that milestone.

Adoption and dissemination will run on the engine that already built FreeMoCap. The standing collaborator network and community described above are the distribution channel: every validated DOME instance at a partner site is both a scientific node and a reference build others reproduce from a published parts list, build guide, and calibration protocol. Because the platform is grant-funded and we can sell our hardware with very low margins to a reach price point that ensure maximum penetration across the widest possible slice of society. The open-source core invites downstream products, services, and integrations to form around one shared measurement standard, built without black-boxes or IP-induced cones of silence, so the X-Lab's reach is counted not only in units sold but in the independent instruments, companies, and published results that adopt the shared philosophy organized around the love of measurement, and the expansive drive to see more and more fully than we ever could before.

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4. Cherian, A. T. Open-Source Development and Validation of a Low-Cost Markerless System for Quantitative Motion Analysis. (Boston, Massachusetts, 2026).
5. Muller, K. S. *et al.* Foothold Selection during Locomotion in Uneven Terrain: Results from the Integration of Eye Tracking, Motion Capture, and Photogrammetry. <https://elifesciences.org/reviewed-preprints/91243v2> (2024) doi:10.7554/eLife.91243.2.